Murtaza Afzali

Full-Stack AI Engineer · LangGraph Multi-Agent Systems · Production RAG

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PROFILE

AI engineer who designs and ships production-grade multi-agent systems: LangGraph orchestration, anti-hallucination RAG pipelines, and real-time Next.js / FastAPI products. Builds for reliability, with checkpointing, smart retries, cost control and per-node observability, so systems keep working outside the demo.

SELECTED PROJECTS

VidBrain — AI Video Intelligence — Flagship · BSc thesis

LangGraph · Pinecone · Supabase · FastAPI · Next.js · SSE

- Built an 8-node self-correcting pipeline (supervisor, reranker, validator, web fallback) that answers from the video only and labels web-sourced answers.
- Small-to-Big retrieval: small chunks in Pinecone, parent passages in Supabase; cross-encoder reranking with automatic LLM fallback; per-user namespace isolation.
- Self-RAG graders and verified citations: timestamps come from retrieved data, never from model-claimed IDs.
- Single-flight indexing (Postgres lock), dedup cache, transient-vs-permanent retries, checkpoint resume, and a traces dashboard with RBAC.

AI Prescription Safety Copilot — Clinical decision support

LangGraph · Pinecone · Supabase · Next.js

- 11-node graph: extractor, supervisor, parallel safety, lab, contraindication and guideline agents, with a reflection gate and reviewer.
- Deterministic Python dose validation (no LLM arithmetic); fail-closed printing; doctor override only with a documented reason.
- Fast Mode (Extractor + Safety, about 1.5 s) vs Copilot Mode to control cost and latency.

Ariana Feast — Restaurant & Hotel Agent — Live demo

LangGraph · FastAPI · Next.js · Supabase · OpenRouter · Vercel

- Tool-calling agent (7 tools) with human-in-the-loop checkout and room booking; cart lives in the database, so chatbot and website stay in sync in real time.
- Business rules enforced in code, not prompts; race-condition guard recalculates the live total after a HITL pause; per-user daily message cap.

NovaCare — Clinic Appointment Booking — Agents + web app

LangGraph · pytest · Next.js · TypeScript · Supabase Auth · RLS

- Multi-agent booking with HITL confirmation before every database write; full pytest suite.
- Web app with patient / doctor / admin roles, Row Level Security, 7-day slot logic, bilingual fa/en with RTL.

Also built

- **Agentic Telegram Learning Bot:** LangGraph, Pinecone, SqliteSaver, 24 tests that need no API keys.
- **Agentic RAG Router:** routes between Pinecone PDF search and live Tavily web search.
- **Portfolio site** with a RAG chatbot, plus a Finances dashboard (Next.js, shadcn/ui).

CORE SKILLS

AI & Agents

LangGraph · LangChain · Multi-agent · HITL · Checkpointing · Structured output

RAG & Retrieval

Pinecone · Small-to-Big · Reranking · CRAG · Self-RAG · Tavily

Backend

Python · FastAPI · Async · SSE streaming · SQLModel · pytest

Frontend

Next.js · React · TypeScript · Tailwind · shadcn/ui · Framer Motion

Data & Deploy

PostgreSQL · Supabase · RLS · Docker · Vercel · Git / GitHub

LLM APIs

OpenAI · Anthropic · OpenRouter

PRODUCTION ENGINEERING

- Cost control: dedup, caching, context trimming
- Transient vs permanent error retries
- Guardrails and anti-hallucination gates
- Tracing, monitoring dashboards, RBAC
- Tenant isolation and race-condition safety

EDUCATION

BSc Computer Science (Software)

Herat University · 2022 – Aug 2026

Thesis: VidBrain, a self-correcting multi-agent RAG system for video.

LANGUAGES

Dari / Persian — native

English — professional working (technical writing, documentation)

LINKS

Portfolio
ai-portfolio-monorepo.vercel.app

GitHub
github.com/MurtazaAfzali13

LinkedIn
linkedin.com/in/murtaza-afzali-484446405

Live demo
autonomous-restaurant-agent.vercel.app